Thus reducing the size of equipment considerably. Let we see in detail about this much important semiconductors.

It is not easy to define a semiconductor, if we want to take into account all its physical characteristics. However, generally a semiconductor is defined on the basis of electrical conductivity as under. A semiconductor is a substance which has resistivity (10-4 Ωto 0.5Ω) between conductors and insulators, e.g. Germanium, Silicon, Selenium, Carbon, Graphene, etc. Table 3.1 shows the resistivity of various semiconducting materials.

Table 3.1 Resistivity of Semiconductor Materials

Sl. No.	Substance	Nature	Resistivity
1.	Copper	good conductor	1.7×10^{-8} Ωm
2.	Germanium	semiconductor	0.6 Ωm
3.	Glass	insulator	9×10^{11} Ωm
4.	Nichrome	resistance material	10^{-4} Ωm

3.6.1 Properties of Semiconductors

The resistivity of a semiconductor is less than an insulator but more than a conductor.

Semiconductors have negative temperature co-efficient of resistance, i.e., the resistance of the semiconductor decreases with the increase in temperature and vice-versa. For example, germanium is usually an insulator at low temperature but it becomes good conductor at high temperature.

When metallic-impurity (e.g. Arsenic, Gallium, etc.) material is added to the semiconductor material then the current conduction property of the material changes, appreciably. This property is most important and is discussed later in detail.

3.6.2 Bonds in Semiconductors

In semiconductors, bonds are formed by sharing of valence electrons. Such bonds are called covalent bond. In the formation of a covalent bond, each atom contributes equal number of valence electrons and the contributed electrons are shared by the atoms engaged in the formation of the bond. The covalent bond of Germanium is shown in Figure 3.17.

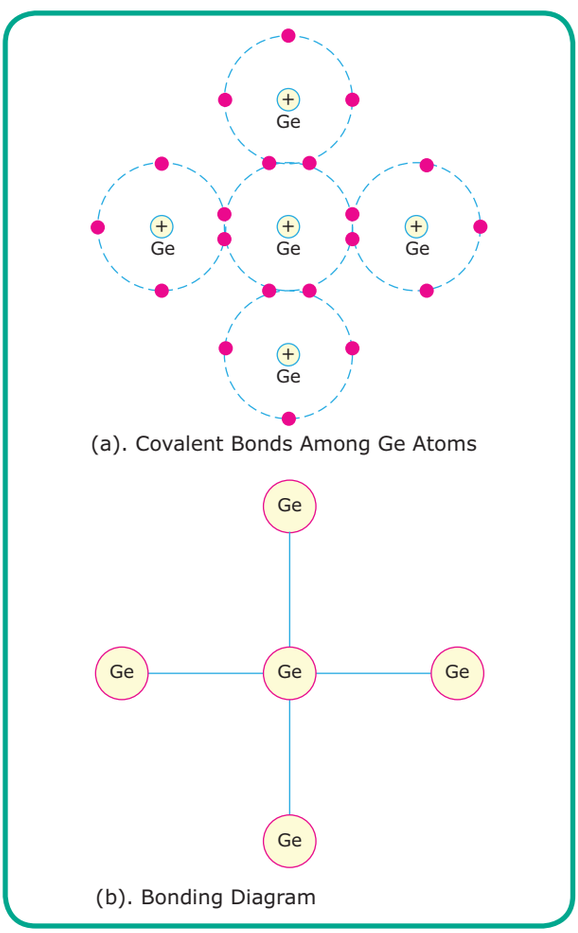


Figure 3.17 Formation of covalent bond

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